

Institute of Actuaries of India

Subject ST8 – General Insurance: Pricing

October 2015 Examination

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

a) **To constrain selected rating factors** – due to regulatory or marketing considerations, we may have to constrain certain rating factors to take on pre-specified values. E.g., discount provided on personal motor insurance for different levels of No Claim Discount.

We must fix the values of certain rating factors in advance, and allow the remaining rating factors to optimally conform to these constraints.

b) Offset can be **used in sequential modeling** – this can be applied while building a credit scoring model on top of an existing rating plan.

This is achieved by fitting the first model, offsetting all the values (*ie* fixing the parameters at the values obtained in fitting the first model), and then fitting a second model to explain the remaining patterns in the data.

c) **To compare a final model to a new “hold-out” sample of data** - to do this, first divide the data into modelling and test sets. A division like $\frac{3}{4}$ and $\frac{1}{4}$ may be appropriate. Then fit the model on the main dataset, and then fully offset it. Offsetting fixes the parameters so that they can be reloaded and scored against the test set.

[4]

Solution 2:

- i) Product recall insurance is usually triggered by any of the following insured events:
- a. Accidental product contamination which occurs during manufacture, packaging, storage or distribution.
 - b. Malicious product tampering of the insured product
 - c. Product extortion demand – threats received by the insured to commit malicious product tampering
 - d. Mislabelling regarding the instruction to use the product

[2]

- ii) Coverages are generally available against three categories of cost incurred – pre-recall costs; recall costs and post-recall costs.

Pre-recall costs are comprised of identification and analyses of potential cause, product testing at laboratories, recall and crisis management plan testing.

Recall costs are incurred in the areas of media and public relation, transport and logistics, storage, replacement, redistribution, staff overtime etc.

Post-recall costs are comprised of cleaning cost, loss of gross income, repair cost and rehabilitation cost.

Third party expenses can also be covered for retailers who have to remove the affected products from their shelves and lose business.

[3]

- iii) Product recall insurance is usually a first-party coverage for the insured's economic losses resulting from product recalls. On the other hand, product liability insurance is coverage for third-party liability claims for physical damage or injury arising from defects or problems related to products.

Under product recall insurance, the insurer reimburses various expenses, even if the product has yet to cause an actual liability and may never result in such litigation. The trigger for the insurance is the insured event— not the recall itself. For instance, if a product is learned to be defective and has yet to leave the manufacturer's premises or enter the marketplace, the insurance is automatically triggered.

[2]

[7 Marks]

Solution 3:

- i) In the sliding scale commission, the commission percentage, varies based on the actual loss experience whereas it is fixed in the profit commission.

This is particularly suitable where both the insurer and reinsurer view the business as profitable but their expectations on the loss ratio or the extent of profit differ.

Also, since the loss ratio is a random variable, the commission to be paid out by the reinsurer will be lesser in case of bad experience while greater in the case of a good year.

Due to this loss sensitive feature a certain portion of risk, extent of which depends on the sliding commission structure, is passed back by the reinsurer to the insurer.

[3]

- ii)

This depends on the amount of expected losses under the XoL layer. If the expected losses are low, relatively speaking, then sliding scale or profit commission are not suitable.

In this scenario, it won't be suitable for the higher layer but may be suitable for the working layer

[2]

iii)

Treat the day before day 1 + day 1 + day 2 as one event. RI recoverable = $\min(10+8 - 5, 20) = 13$

Treat day 3 + day 4 + day 5 as another event. RI recoverable = $\min(15+12 - 5, 20) = 20$.

Total RI recoverable = 33

[2]

[7 Marks]

Solution 4:

- i) To be more successful than its competitors would motivate an insurer to become more refined in its risk classification system and thus its pricing structure, so that it could serve both lower cost and higher cost risks in the marketplace.

Thus, there is an incentive for risk classifications, as used by competitive insurance programs, to become more refined and to more accurately reflect the differentials in expected costs among identifiable classes of risks.

Economic incentive also requires the risk classification system to be efficient. The additional expense of obtaining more refinement should not be greater than the reduction in expected costs for the lower cost risk classification. Thus, there is a practical limit to the incentive to add refinements to the classification system.

In general, economic incentive operates over time to favor classification systems that result in a price for each risk which most nearly equals the expected cost associated with the class to which that risk is assigned. [3]

- ii) Consumer Price Index (CPI) at the component level (e.g., medical cost and construction cost indices): To find trends in expenses, premium, or losses etc. that are relevant to the insurance product being priced.

Fire Station Database: This will help identify the distance / travel time to the nearest fire station as this could be an important factor that determines the magnitude of loss.

Natural Perils Information (e.g., average rainfall or number of days below freezing of a given area) / Event Module by Location for Natural Perils used in CAT Models: This could help in risk classification for property lines, Motor Insurance etc. For example, high rainfall or foggy conditions could lead to more Motor Own Damage claims.

Credit Scores from a Credit Rating Agency: For personal lines, credit score of the individual is thought to be an important predictor of risk. However, acceptability by law and also by the market needs to be considered.

Geo-demographic data (i.e., average characteristics of a particular area) such as census data: For example, population density can be an important predictor of the frequency of motor accidents.

Crime Data by Region such as Theft indices: This could be an important predictor of frequency of vehicle thefts in motor Insurance or for Burglary or Homeowners Insurance.

Financial Reporting and Statutory Filings of Publicly Listed Companies: This could give insights into the risk profile of the industry that the corporate operates in, with regards to Directors & Officers Insurance.

Other examples include Traffic violations data, Road accidents database, Telematics etc.

[5]

[8 Marks]

Solution 5:

- i) For some books of business there can be a low persistency rate and “new” business may comprise a significant part of the book. Or a growing book where new business is considerably high proportion every year. For these lines, more detailed analysis is required.

If the underwriters offered slightly lower premium rates or better terms to attract the new business, the actual rate change might be lower than what was stated. Or if they actually charged more for the new business, the actual rate change might be greater than what was stated.

If the accounts that were not renewed were more profitable when compared to the renewed business, the actual rate change might be lower than what was stated.

Need to gain understanding of how the rate changes were calculated for individual renewals as there are several ‘soft’ factors involved. For example, how changes to coverage or policy terms or the risk are considered. **[3]**

- ii) Depending on how many years of history the cedant is providing, it may be excessive to allow for a loss of that size to occur once in that many years. The reinsurer could instead remove the loss, do the loss cost analysis without it, and then add a loading for large losses at the end.

The reinsurer could base this loading on an analysis of the reinsurer’s entire book, or allow for it by spreading the cedant’s own large loss over a more reasonable period of time.

- iii) Claims for which the reported amounts are lower than threshold but whose trended amounts, at the current levels, exceed the threshold will not be available. These claims, if they were to happen today, are likely to hit the reinsurance layer and hence are to be considered.

Volume of claims data now obtained for different years will now be consistent. Otherwise, most of the claims provided would be from the recent years and very few from older years [2]

- iv) Two different risks or portfolios may have the same mean and standard deviation but the loss distribution could be very different. In such cases, using the same proportion of standard deviation may not be appropriate. [1]

[8 Marks]

Solution 6:

- i) Average claim amount with limit at 100000 =
 Total amount of paid w.r.t claims with limit 100000 / Total number of claims with limit 100000 /
 $= 68455808/1136 = 60260$

Similarly, average claim amount with limit 250000 = 85074, while
 average claim amount with limit 500000 = 85057.

Since average claim size for 500000 on this empirical data is less than the average claim size for 250000, the resultant ILF for 500000 would be less than 250000.

This is happening due to random variation as the number of claims below 100000 for limit 500000 is quite low. Even for 250000, the number is not statistically large enough.

[3]

- ii)

To find the average claim amount for the layer 0 to 100000, claims with all per-claim limits will be considered. To find the average claim amount for the layer 100000 to 250000, claims with per-claim limit of 100000 will not be considered. Similarly, for the layer 250000 to 500000, only claims with per-claim limit of 500000 will be considered.

Table 1

Claim Size	Per-Claim	Number of	Amount of Claims	Amount in Layer 0 to	Amount in Layer 100000	Amount in Layer
0-99999	100000	810	35,855,808	35,855,808	Not	Not
0-99999	250000	394	17,800,556	17,800,556	0	Not

0-99999	500000	208	8,732,804	8,732,804	0	0
100000	100000	326	32,600,000	32,600,000	Not Considered	Not Considered
100001-249999	250000	135	20,276,866	135 * 100000 = 20,276,866	20,276,866 – 135*100,000	Not Considered
100001-249999	500000	68	10,138,725	68 * 100000 = 10,138,725	10,138,725 – 68*100,000 =	0
250000	250000	42	10,500,000	42 * 100000 = 10,500,000	42 * (250000 – 100000) =	Not Considered
250001-499999	500000	14	4,550,252	14 * 100000 = 4,550,252	14 * (250000 – 100000) =	4,550,252 – 14*250,000
500000	500000	3	1,500,000	3 * 100000 = 300,000	3 * (250000 – 100000) =	3 * (500,000 –

Table 2

Layer	Total Amount in the Layer	Number of Claims considered in the	Average Severity for
0 to 100000	Sum of (c) from Table 1 =	Sum of (a) from Table 1 = 2000	60,594.58
100000 to 250000	Sum of (d) from Table 1 =	Sum of (a) from Table 1 = 2000 –	21950.92
250000 to 500000	Sum of (e) from Table 1 =	Sum of (a) from Table 1 = 208 +	6144.20

ILF for Limit 250000 w.r.t base limit of 100000 = $(60595+21951)/60595 = 1.36$

ILF for Limit 500000 w.r.t base limit of 100000 = $(60595+21951+6144)/60595 = 1.46$

[4]

- iii) Even if the proportionate increase in claim amount due to the new ruling is the same across all claim sizes, the ILFs 500000 and 250000, w.r.t base limit 100000, will increase.

Since the new rule impacts only the severe claims, the claims amounts (before per-claim limits) higher than 500000 are more likely to be impacted in terms of proportionate increase compared to claims in band 250000 to 500000. Similarly claims in band 250000 to 500000 are more likely to be impacted compared to claims in band 100000 to 250000. And claims in 100000 to 250000 more than claims in band 0 to 100000.

As a result, ILFs for 250000 and 500000, w.r.t base limit 100000, are expected to increase.

Also the difference between ILFs for 500000 and 250000, w.r.t base limit 100000, is also expected to increase.

[2]

[9 Marks]

Solution 7:

i) The impact that the following factors have on the choice of the weighting between the Pure Experience rated premium and the Exposure rated premium for a particular policy:

- Stability of the claims experience over time: greater the stability, greater that credibility can be attached to the claims experience and hence greater weight to experience rating
- Number of years of available data - a minimum of 3 years of claims data is required with 5 years being preferable for experience rating. Again, more years would lead to greater weight to experience rating.
- Size of the account: Greater the size, greater the statistical credibility and hence greater weighting can be given to being experience rated
- Changes in the type of risks being covered under the policy: This means that the experience is not as relevant and therefore lesser weighting would be attached to the experience.
- Account composition – if there segments of the portfolio that generally experience large infrequent losses then these segments should have a higher weighting towards being exposure rated rather than based on the actual experience
- Consistency of the policy wording over time particularly with regards to the level of excess on the policy or any other terms and conditions: This means that the experience is not as relevant and therefore lesser weighting would be attached to the experience.

[5]

ii) Arguments supporting the use of experience rating in general insurance include:

- premiums better reflect the risk
This is because risks belonging to the same rating group in the exposure rating plan could still be heterogeneous.
- encourages better behaviour by policyholders
which in turn discourages small claims and associated admin costs
- Experience rating might be wanted by policyholders.

[2]

- iii) The quality of the service is a more important form of competition on commercial lines, e.g. a fleet manager with a large fleet will want efficient handling of the many claims likely to be made in each year.

Also, businesses should pay more attention to the financial strength of the insurer to ensure that claims will be paid.

[2]

[9 Marks]

Solution 8:

i) Loss from j^{th} risk, $S_j = N_j X_j$ where:

- $N_j = 1$ with probability p ; 0 otherwise
- X_j follows exponential (PDF: $\frac{1}{\theta} e^{-\frac{x}{\theta}}$) distribution. Mean is θ

$E[S] = E[S_1] + E[S_2] + E[S_3] + \dots + E[S_n] = n \cdot E[S_j]$ (Since, all risks are identical)
 $= n \cdot \text{Probability of a claim} \cdot \text{Expected amount of claim (since amount is independent of number of claims)}$
 $= n \cdot p \cdot \theta$

$\text{Var}[S] = \text{Var}[S_1 + S_2 + S_3 + \dots + S_n] = \text{Var}[S_1] + \text{Var}[S_2] + \dots + \text{Var}[S_n]$ (since all risks are independent)
 $= n \cdot \{ E[S_j^2] - (E[S_j])^2 \} = n \cdot \{ p \cdot m_2 - (p \cdot \theta)^2 \} = np\theta^2(2 - p)$ Where $m_2 = \text{second moment of exponential} = 2\theta^2$ (Since, all risks are identical)

Alternatively:

Treat it as a collective risk model with $N \sim \text{Bin}(n, p)$ and $X \sim \text{Exp}(1/\theta)$. This is possible as all the risks are identical; the individual risk model can be taken as a special case of the collective risk model.

Then, $E[S] = E[N] \cdot E[X] = n \cdot p \cdot \theta$

$\text{Var}[S] = E(N)\text{Var}(X) + \text{Var}(N)[E(X)]^2 = np\theta^2 + np(1 - p)\theta^2 = np\theta^2(2 - p)$

[2]

ii) The variance would be greater than the previous case.

This is because of the parameter uncertainty as the exact values of p and θ are not known with certainty.

The previous variance incorporated only the process uncertainty.

Now, the variance should incorporate both process uncertainty and parameter uncertainty. [2]

iii)

1. We will generate 10000 or 100000 simulations for the random variable S. From the list of 10000 or 100000 generated values for the random variable S, find mean and standard deviation.
2. In the each simulation for S, we perform the following steps: (one mark for each point)
 - a. Generate a random value from continuous uniform distribution $U(0.9\hat{p}, 1.1\hat{p})$. Denote this value as 'r-p'
 - b. Generate a random value for number of claims from binomial distribution $\text{Bin}(n, r-p')$. Denote this value as 'r-N'.
 - c. Generate a random value from continuous uniform distribution $U(0.9\hat{\theta}, 1.1\hat{\theta})$. Denote this value as 'r- θ '
 - d. Generate 'r-N' random values from Exponential distribution $\text{Exp}(1/r-\theta')$
 - e. Take the sum of the values generated from part (d) which gives the generated random value for 'S', the aggregate claims distribution

Alternatively:

- a. Generate a random value from continuous uniform distribution $U(0.9\hat{p}, 1.1\hat{p})$. Denote this value as 'r-p'
- b. Generate a random value from continuous uniform distribution $U(0.9\hat{\theta}, 1.1\hat{\theta})$. Denote this value as 'r- θ '
- c. Use the generated random values 'r-p' and 'r- θ ' to find $E[S]$ and $\text{Var}[S]$, as derived in part (i) above.
- d. Take the average of $E[S]$ and Square Root of $\text{Var}[S]$ values across 10K/100K simulations as the estimates for mean and standard deviation, respectively, of the aggregate claims distribution.

[5]

[9 Marks]

Solution 9:

- i) The following data inputs will be required:
 - a. Current rate and rating factors of the company used to determine premiums
 - b. Proposed rate and rating factors of the company
 - c. Past rate changes and impact on retention

- d. Rates and rating factors of the key competitors
- e. Profitability of different classes of policyholders for the company
- f. Current in-force distribution of policyholders in different rating classes

[2]

- ii) The following data inputs will be required: 9% increase in premium would make the new premium as INR 545. From the table, we can see that 5% of the existing policyholders will shop around when premium increases by INR 45.

When they shop, they will find out that Competitor 2 is charging the lowest premium and saving of INR 93 is possible and 40% of them will switch for this amount of savings.

So the rate increase will cause 2% ($= 5\% \times 40\%$) of the policyholders to leave the company. There are 250 policyholders under this specific set, which means 5 policyholders will leave and go to other company.

These policyholders were expected to earn annual profit of 13.4% at this rate level.

Current premium = INR 125,000 ($= 500 \times 250$)

Premium after rate change = INR 133,525 (545×245)

Profit after rate change = INR 17,892

Similarly for 12% rate increase, 24% of the existing policyholders will possibly leave the company. Hence,

Premium after rate change = INR 106,400 ($= 560 \times 190$)

Profit after rate change = INR 16,704

So, in view of larger total profit, the Company should prefer a rate increase of 9%.

[4]

- iii) The company needs to first define the type of customer it wants to attract and then understand how much price sensitive these customers are. It can then decide the optimal new business pricing strategy at a target level of profitability by analyzing price sensitivity and expected sales volumes at different profit levels.

The objective of profit optimisation process is to find the premium level that will maximize the benefit function.

In order to develop the profit optimisation model for new business, the following information will be required – price elasticity; output from pure premium model; financial objectives of the insurer; cross selling information and competitor's rates.

In this modelling exercise, we need to determine the constraint and the target. For example, we can define total new business revenue and maximize profitability given that revenue; or define the overall new business profitability and maximize new business revenue given that profit target.

To carry out the process, an appropriate software can be used which can analyse all the elasticity of demand functions for potential clients. With this analysis, the company can determine probabilistically the best combination of conversion rates and risk profiles, and then produce the desired maximum number of new business conversions given an overall profit target.

Regulatory restrictions and management constraints can also be incorporated.

The output of this process produces is an efficient frontier where each conversion rate will have a combination of policies and prices that produces different profitability levels. However, only one combination of policies and prices will give the maximum profitability for a given overall conversion rate.

[3]

[9 Marks]

Solution 10:

- i) Claim inflation impacts the home insurance claims cost on its content and building cover separately.

Household contents insurance

The most commonly quoted inflation indices are commonly known as retail-price inflation or consumer-price inflation. These may reasonably be expected to affect the cost of claims of household contents policies, although they may not be an entirely accurate predictor of claims costs since consumer goods are likely to be the subject of claims whereas the inflation indices may include items such as food and housing costs.

Buildings insurance

Private, commercial and industrial property claims will generally be for the repair and rebuilding of property, particularly buildings. These costs will be linked to the cost of building materials, but labour will be a major part of the claims costs. This cost is more likely to be linked to wages than prices; wages tend to increase more than prices so the cost of this insurance is likely to increase more than general prices.

Another aspect of inflation is “loss amplification” or “demand surge”. This arises where there is a temporary increase in costs of labour or raw materials due to a large number of claims being made at the same time, for example, as a result of a single large catastrophic event.

[3]

ii) The basic structure of a catastrophe model is comprised of the following interlinked modules –

- Event module
- Hazard module
- Inventory or exposure module
- Vulnerability module and
- Financial analysis module

A stochastic event set should be created using past experience and a scientific understanding of the underlying causes of earthquake. It may also include earthquake that may have not happened at past.

For each of the earthquakes within the catastrophe model event set, the model will include a number of parameters such as moment magnitude; focal depth; total area of fault rupture and fault type.

Event frequency may be modelled using a statistical distribution, such as a Poisson.

The hazard module will specify that if earthquake happens again, then it is likely to result in some ground shaking.

This will cause building damage and collapse; sprinkler leakage can cause extensive damage to the contents.

The extent of damage from shaking depends on the amplitude, duration, and frequency content of the ground motions. Shaking intensity will be calculated at this stage.

Under inventory module, we need to input detailed exposure database of the insured buildings, location, age, occupancy and type of construction.

The inventory module will also contain the values of the buildings and contents that are to be insured.

The vulnerability module produces the modelled loss based on the values of the buildings and contents that are to be insured, not the actual insured limits.

This module describes the interaction between buildings, including their structural components as well as their contents, and the shaking intensity to which they are exposed.

Financial analysis module will convert the physical damage loss from vulnerability module to gross insured loss depending upon the policy conditions.

We need to input data related to policy limits and sub-limits, excess, coverage terms etc.

This module may also apply various types of reinsurance purchased to protect the portfolio.

[5]

iii) Underwriting & Product Pricing

- Risk selection and pricing at the point of sale
- Setting underwriting guidelines
- Use model outputs in rate filings and in pricing of individual policies or programs

Reinsurance Purchase

- To structure reinsurance program and evaluate appropriate level of reinsurance purchase
- Streamline efficiency of communication with reinsurance intermediaries

Claims Management

- Reserving assessment of major catastrophic events
- Advance planning and resource deployment
- Post event communications

Enterprise Risk Management

- To monitor aggregate insured losses as part of risk management
- Communicate with rating agencies
- Capital allocation and assessment, for both internal and regulatory purposes
- Manage the impact of catastrophe risk on surplus

Business Planning

- To create disaster recovery plans
- Financial planning

Portfolio Optimisation

- Identification of areas to grow or retract based on model based risk metrics
- Perform model based analyses to understand and manage the drivers of catastrophe risk

Investment

- Designing insurance-linked derivatives such as catastrophe bonds.

[6]**[14 Marks]****Solution 11:**

- i) Through internet – online quote generation engine and policy issuance

Through tied agents like banks

Through motor dealers

Direct marketing through insurer's own staff (cold-call selling by post or telephone)

[1]

- ii) The number of written policies divided by the number of quoted policies in a given period

[1]

iii) Data

- Since the company is trying to understand competitor's tariff structure for the business sold through internet, the data need to be collected for this exercise should be in the form of insurance quotations obtained through the online quote engine of the chosen competitors. The quotes can also be collected from web aggregators.
- Apart from the quotes, we need to understand the product features of the competitors' so that like with like comparison can be made.

Data collection

- Since in the comprehensive motor insurance a large number of rating factors are usually involved, the exhaustive combination of all these factors and their underlying levels will lead to millions of risk profiles. This will be impossible to collect and manage properly.

- Depending upon the company's profile mix for past two years, a representative quote basket needs to be created first. It should be concentrated on those risk profiles, to which the company sold the product in past and also intended to sell in future. Thus the number of possible combinations can be reduced significantly.
- Profiles should be chosen to represent the range of different rating factor values and the profiles should be weighted to represent their relative importance.
- Depending upon the number of competitors in the market, the company may need to restrict the exercise for only the key 2 or 3 competitors.
- For each of these competitors, using the risk profiles in the quote basket, quotes need to be collected online within a stipulated timeframe.
- Quotes can be collected manually or using any web crawler.

Challenges

- While collecting the quotes from competitor's website, they may identify this as a large number of almost similar profiles are used in the process. There is a possibility that the IP address can be easily recognized and subsequently blocked, thereby no option of further quote collection.
- Due to the dynamic nature of motor insurance business, competitors can revise their premiums frequently. If such price revision takes place while collecting the quotes, then a part of the data collected before the revision may need to be collected again to avoid inaccuracy.
- It's a very time consuming process and will require significant man power, in case manual quote collection.
- The competitors' premiums that are collected are office premiums and not the pure premiums.

[5]

- iv) Once the quote basket is ready and the quotes are being collected, some preliminary testing needs to be done and on the basis of the results the quote basket can be refined for appropriate data collection.

Keep those rating factor values in the analysis that have regard for sensitivity testing of the rating factor values being used. The profiles need to include more samples for the rating factors that are flagged by the sensitivity testing.

Driver related factors:

There is expected to be a large number of driver related questions being asked by competitors' online quote engine to generate quotations. E.g., age, ownership, past driving convictions, license age, number of additional driver, vehicle financing, employment status, purpose of use, NCD history, claims history etc.

A base profile can be created upfront at random, or may be by paying emphasis to a key risk profile for which the company has sold the highest number of policies. Each driver related question from the list (excluding age) should then be put under test conditions, where all the other rating factors will be fixed at base, and the rating factor under consideration is varied for different levels.

First cut relativities can be obtained through this one-way analysis.

But there may be possible interactions with other rating factors. Hence the relativities obtained at the initial stage need to be put under further test conditions in a similar way.

Vehicle related factors:

This part is very complex and the difficulty increases due to a very high number of different individual vehicles plying on the road. The key challenge will be to classify the vehicles into homogeneous groups of similar risk.

A possible approach can be based on segmenting the portfolio according to the combinations of vehicle characteristics, e.g., power to weight ratio. As this is a large insurer, we should not encounter problem in finding suitable exposure on different segments.

The following vehicle characteristics can be used in the process – make and model; power to weight ratio; vehicle value; year of manufacture; engine size etc.

GLM can be used to find out the relativities of different rating factors by considering the quoted premium as response variable and the other factors related to vehicle characteristics as explanatory variables.

Suitable distribution and link function need to be used while fitting GLM on the collected quotes.

We can keep the other rating factors (driver and location related) fixed and varying the vehicle characteristics for the base risk profile (identified earlier) to create the quote dataset for GLM exercise.

Location related factors:

Rates across locations can be deconstructed using area codes, e.g. post code.

One way to find out location relativities is to carry out one-way analysis as we have done in case of driver related factors.

Here, we need to find out the premiums for different locations against the same risk profile. This should be considered while designing the quote basket.

In order to find out interaction effect of other rating factors, we can create multiple risk profiles and compare the premiums for those at different locations. In case there is no change in premium due to risk profile, we may conclude that location behaves independently.

Otherwise, a mathematical function can be obtained to explain the relationship of location related factors (such as socio-economic index) and the quoted premium. This can be done using a sample of locations at the beginning and then interpolate for the remaining locations.

[6]

- v) Insurance cycle is the observed tendency of insurance prices and hence profitability to vary over a period of several years.

The cyclical nature of premium rate movements is a very challenging aspect of managing a portfolio of risks. It is important that the insurer is aware of the stage of the insurance cycle that the market is in when deciding on the final premium.

There may be different reasons causing insurance cycle such as –

- A delay in the understanding of the emergence of higher claims costs, which leads to under-pricing of current risks.

- Inflows and outflows of capacity to the market, often as a result of large-scale catastrophes. Catastrophes may jolt prices higher due to providers leaving the market hence causing a reduction in supply.
- Deliberate under-pricing by key market players in an attempt to drive out competitors.
- An attempt to grow in volume in order to cover high fixed expenses (or to achieve market share).
- Pricing strategy being determined by chasing market prices (upwards or downwards) rather than being based on sound technical prices, with no player in a given market willing to be the first to break the cycle.

[3]

[16 Marks]
